IMDA OPEN INNOVATION PLATFORM · CALL 28 · NTUC LEARNINGHUB

CareerFlow

End-to-End Career Placement Programme Management Platform

One integrated platform that automates the full trainee lifecycle — from enrolment through verified employment and SSG compliance reporting — purpose-built for WSG/SSG-regulated workforce development.

Submitted by **Humankind Design Pte Ltd** · Singapore · April 2026

THE PROBLEM

Placement workflows run on Excel, SharePoint, and memory.

NTUC LHub's programme coordinators manage **200+ trainees** across multiple concurrent WSG/SSG programmes using fragmented CRM, SharePoint, and spreadsheet tools. Four specific pain points recur:

- Subjective, salesperson-led course matching with no standardisation
- Manual document chasing for employment verification evidence
- Monthly SSG compliance reports compiled by hand
- At-risk placements spotted only after they've stalled

IMPACT TARGETS

+10–30% Placement rates	**–30–50%** Admin time
100% SSG reporting automated	**1** Platform replacing 3+ tools

THE SOLUTION

Four connected modules. One coherent workflow.

01 Smart Enrolment

AI assessment analyses trainee backgrounds, identifies transferable skills, and recommends optimal courses with confidence scores. Replaces subjective evaluation with standardised, defensible decisions.

02 Trainee Tracker

Real-time pipeline from Application → Enrolled → Training → Placed → Verified, with automated document reminders and trainee self-service uploads for employment evidence.

03 Placement Engine

Skills-to-job matching ranked by fit score, with proactive at-risk alerts when placements stall or employer engagement drops. Shifts the model from reactive tracking to active prevention.

04 Analytics & Compliance

Live KPI dashboard, automated monthly SSG compliance reports, PDPA consent management, and full audit trails — compliance woven in, not bolted on.

Working Prototype — Live Today

A functional prototype demonstrates all four modules through scenarios drawn directly from the challenge brief.

https://juhus.github.io/ntuc-lh-careerflow/

Sarah — Programme Coordinator managing pipeline, dashboard, and at-risk alerts

Marcus — Career Switcher through AI-assisted course matching and skills-gap analysis

Automated Compliance — SSG report generation, audit trail, PDPA consent

WHAT SETS US APART

Purpose-built for WSG/SSG programmes — not a repurposed CRM.

Domain-Native Design

Understands the full trainee lifecycle natively. Compliance, audit trails, and PDPA consent woven into every module.

AI-Powered Consistency

Confidence-scored recommendations replace subjective judgment with standardised, defensible decisions at scale.

Proactive Intervention

Detects stalled pipelines, dropped employer engagement, outdated CVs — prevention, not post-mortem.

Integration-First

Coexists with HubSpot, Dynamics 365, SharePoint, and Excel. Progressive automation, not rip-and-replace.

Human-Centric Design

Built by a design consultancy specialising in enterprise workflow. Minimal training overhead, maximum adoption.

Cloud & PDPA-Ready

AWS-native architecture aligned with NTUC LHub's environment. Encryption, RBAC, audit logging as defaults.

THE TEAM

30+ combined years building enterprise management platforms.

Soh Ju Hu

STRATEGIST & PROJECT LEAD

10 years in workflow design and digital product development across logistics, F&B, and retail. Owns solution architecture, client engagement, and delivery oversight.

Jemima Song

LEAD DESIGNER

10 years in UI/UX and enterprise product design. Balances information density with usability for coordinators managing hundreds of trainees. Owns design system and accessibility.

Tung Nguyen

CHIEF TECHNOLOGY OFFICER

10 years in full-stack engineering and AI/ML integration. Deep CRM/SharePoint/reporting API experience. Owns technical architecture, AWS infrastructure, and PDPA data security.

DELIVERY PLAN

16 weeks from discovery to pilot deployment.

WEEKS 1–3

Discovery & Design

- Stakeholder interviews
- Workflow mapping
- UX sprints

WEEKS 4–8

Core Build

- Smart Enrolment + AI
- Trainee Tracker
- CRM / SharePoint integration

WEEKS 9–11

Matching & Reporting

- Placement Engine
- Analytics dashboard
- SSG, PDPA, audit

WEEKS 12–14

Testing & Refinement

- UAT with NTUC LHub
- Performance & security
- Training materials

WEEKS 15–16

Pilot Deployment

- One programme cohort
- Feedback & iteration
- Handover & support

SCALABILITY & COST

SaaS economics. Government-grade architecture.

Architecture

- → AWS cloud-native — aligned with NTUC LHub's existing infrastructure
- → Microservices allow independent scaling per module
- → API-first design for future integration with additional government systems

Cost Model

- → SaaS subscription — no upfront infrastructure investment
- → Pricing scales with programmes/trainees, not per-user
- → **SGD 3,000–8,000/month** at typical scale, below combined cost of current tooling + manual staff time

Growth Path

- → Expandable to other training providers, polytechnics, and government agencies running career transition programmes
- → Applicable beyond ICT — healthcare, finance, logistics, green economy, any sector tracking career placements
- → API marketplace for employer integration — job posting sync, interview scheduling

Readiness

- → Prototype / Proof of Concept · Easily Scalable
- → Team ready to work on-site with NTUC LHub in Singapore